TOTAL COST OF OWNERSHIP (TCO)

COMPREHENSIVE FINANCIAL ANALYSIS REPORT

Project: t1

Generated: December 27, 2025

EXECUTIVE SUMMARY

Initial Investment: €1,151,000.00

Annual Operating Costs: €593,200.00

5-Year Total Cost: €4,117,000.00

ROI: 3.8%

Payback Period: 2.6 years

1. INITIAL INVESTMENT BREAKDOWN

Equipment Purchase:	**€580,000.00**
Installation:	**€116,000.00**
Infrastructure:	**€400,000.00**
Training:	**€30,000.00**
Initial Consulting:	**€25,000.00**
TOTAL INITIAL INVESTMENT	**€1,151,000.00**

2. ANNUAL OPERATING COSTS

Energy:	**€50,000.00**
Maintenance:	**€23,200.00**
Labor:	**€340,000.00**
Insurance:	**€15,000.00**
Consumables:	**€120,000.00**
Utilities:	**€45,000.00**

TOTAL ANNUAL OPERATING COSTS	**€593,200.00**

3. ASSET-BY-ASSET COST BREAKDOWN

High-Shear Reconstitution Tank

Supplier: High-Shear Reconstitution Tank Supplier 2

Purchase Price: €100,000.00

Installation: €20,000.00

Annual Maintenance: €4,000.00

Annual Energy: €12,000.00

Steam Cooker-Stretcher (Filatrice)

Supplier: Steam Cooker-Stretcher (Filatrice) Supplier 2

Purchase Price: €220,000.00

Installation: €44,000.00

Annual Maintenance: €8,800.00

Annual Energy: €18,000.00

Automatic Ball Molder

Supplier: Automatic Ball Molder Supplier 2

Purchase Price: €85,000.00

Installation: €17,000.00

Annual Maintenance: €3,400.00

Annual Energy: €8,000.00

VFFS Liquid Packaging Line

Supplier: VFFS Liquid Packaging Line Supplier 2

Purchase Price: €175,000.00

Installation: €35,000.00

Annual Maintenance: €7,000.00

Annual Energy: €12,000.00

4. FINANCIAL METRICS

5-Year Total Cost:	**€4,117,000.00**
10-Year Total Cost:	**€7,083,000.00**
NPV (5 years, 10% discount):	**€568,400.00**
NPV (10 years, 10% discount):	**€1,945,000.00**
IRR:	**0.32%**
ROI:	**3.75%**
Payback Period:	**€2.60years**

5. 10-YEAR FINANCIAL PROJECTIONS

Year	Revenue	Costs	Net Cash Flow	Cumulative
0	€0	€1,151,000	€-1,151,000	€-1,151,000
1	€2,800,000	€2,350,000	€450,000	€-701,000
2	€3,000,000	€2,450,000	€550,000	€-151,000
3	€3,200,000	€2,550,000	€650,000	€499,000
4	€3,400,000	€2,650,000	€750,000	€1,249,000
5	€3,600,000	€2,750,000	€850,000	€2,099,000
6	€3,800,000	€2,850,000	€950,000	€3,049,000
7	€4,000,000	€2,950,000	€1,050,000	€4,099,000
8	€4,200,000	€3,050,000	€1,150,000	€5,249,000
9	€4,400,000	€3,150,000	€1,250,000	€6,499,000
10	€4,600,000	€3,250,000	€1,350,000	€7,849,000

6. RISK ANALYSIS

High Risk Factors:

• Volatility in raw milk powder and fat prices

• Energy price fluctuations (Steam/Gas)

• Food safety and regulatory compliance risks

• Equipment downtime in a semi-automated line

Mitigation Strategies:

• Establish long-term fixed-price contracts with ingredient suppliers

• Install energy recovery systems on the steam cooker

• Implement a robust Preventive Maintenance (PM) schedule

• Invest in automated monitoring systems for HACCP compliance

Sensitivity Analysis:

• 15% increase in raw material costs: Net annual cash flow reduction of $250,000; Payback period extends to 3.8 years.

• 20% increase in energy utility rates: Annual operating cost increase of $10,000; minimal impact on IRR.

7. CONCLUSION

This comprehensive TCO analysis shows a total investment of €1,151,000 with annual operating costs of €593,200. The project demonstrates a 3.8% ROI with a payback period of 2.6 years. The financial projections indicate positive cash flow and sustainable growth over the 10-year period.